Stochastics - KvS - ...

Conditional Expectations

Let $(\Omega, \mathcal{F}, P)$ be a probability space and let $A \in \mathcal{F}$ be such that $P(A) > 0$. Then, $P(\cdot \mid A): \mathcal{F} \ni B \mapsto P(B|A)$ is a probability and we denote by $E[\cdot \mid A]$ its expectation

eg:

3.65:

Let $\Omega = \{1, .., 6\}$, $P(M) = \frac{|M|}{6}$, for $M \subseteq \Omega$, $X: \Omega \ni \omega \mapsto \omega$ and $A = \{2, 4, 6\}$. Since

$$P(\{\omega\} \mid A) = \begin{cases} 1/3 & , \omega \in \{2, 4, 6\} \\ 0 & , \omega \in \{1, 3, 5\} \end{cases}$$

we obtain

$$E[X \mid A] = \sum_{i=1}^{6} i \cdot P(\{i\} \mid A) = 4$$

3.66:

For a random variable X and $A \in \mathcal{F}$ such that $E[|X| 1_A] < \infty$ and $P(A) > 0$,

$$E[X \mid A] = \frac{E[X 1_A]}{P(A)}$$

3.67:

Let X be a random variable such that $E[|X|] < \infty$ and let $(B_i)_{i \in I}$ be an at most countable collection of sets from $\mathcal{F}$ such that $\bigcup_{i \in I} B_i = \Omega$, $B_j \cap B_k = \emptyset$ for $j, k \in I$ with $j \neq k$ and $P(B_i) > 0$ for $i \in I$

Then,

$$E[X] = \sum_{i \in I} P(B_i) E[X \mid B_i]$$

Stochastics - KVB

→ Sums of independent random variables

↝) Convolution of integer valued probability distributions

Let X & Y be independent integer valued random variables.

Then, for $m \in \mathbb{Z}$,

$$P(X+Y=m) = \sum_{k=-\infty}^{\infty} P(X=m-k)P(Y=k)$$

→ eg.

For independent $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ with $\lambda, \mu > 0$,

$m \in \mathbb{Z}$

$$P(X+Y=m) = \sum_{k=-\infty}^{\infty} \underbrace{P(X=m-k)P(Y=k)}_{=0 \text{ if } m-k<0 \text{ or } k<0}$$

Thus for $m<0$

$$P(X+Y=m)=0$$

and for $m \geq 0$,

$$P(X+Y=m) = \sum_{k=0}^{m} P(X=m-k)P(Y=k) = \sum_{k=0}^{m} \frac{\lambda^{m-k}}{(m-k)!} e^{-\lambda} \frac{\mu^k}{k!} e^{-\mu}$$

$$= \frac{e^{-(\lambda+\mu)}}{m!} \sum_{k=0}^{m} \frac{m!}{k!(m-k)!} \lambda^{m-k} \mu^k$$

$$= \frac{e^{-(\lambda+\mu)}}{m!} \sum_{k=0}^{m} \binom{m}{k} \lambda^{m-k} \mu^k = \frac{e^{-(\lambda+\mu)}}{m!} (\lambda+\mu)^m = \frac{(\lambda+\mu)^m}{m!} e^{-(\lambda+\mu)}$$

→ $X+Y \sim \text{Poisson}(\lambda+\mu)$

□

Stochastik - KW 5

Convolution of densities

convolution of $f * g$

For two densities $f, g : \mathbb{R} \to \mathbb{R}$ define $f * g : \mathbb{R} \to \mathbb{R}$ as

$$(f * g)(t) = \int_{-\infty}^{\infty} f(t-s)\, g(s)\, ds \quad \text{for } t \in \mathbb{R}$$

4.4:

Let X and Y be independent random variables with densities f_X & f_Y. Then the density of f_{X+Y} of $X+Y$ is given:

$$f_{X+Y} = (f_X * f_Y)$$

eg: 4.5:

Let $N_1, N_2 \sim \text{Normal}((0,1))$ be independent by Theorem 4.4

$N_1 + N_2$ has the density

$$f_{N_1+N_2}(t) = (f_{N_1} * f_{N_2})(t) = \int_{-\infty}^{\infty} f_{N_1}(t-s) f_{N_2}(s)\, ds$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{(t-s)^2}{2}} e^{-s^2/2}\, ds$$

$$= \frac{1}{2\pi} e^{-t^2/4} \int_{-\infty}^{\infty} e^{\frac{-t^2/2 - 2st + 2s^2}{2}}\, ds$$

$$= \frac{1}{2\pi} e^{-t^2/4} \int_{-\infty}^{\infty} e^{-\frac{(t/\sqrt{2} - \sqrt{2}s)^2}{2}}\, ds \qquad u = t/\sqrt{2} - \sqrt{2}s$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{2}} e^{-t^2/4} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\, du = \frac{1}{\sqrt{2\pi}\sqrt{2}} e^{-t^2/4}, \text{ for } t \in \mathbb{R}$$

$$\Rightarrow N_1 + N_2 \sim \text{Normal}((0, \sqrt{2}))$$

Stochastics IKVB

-> Moment generating function

The moment generating function M_X of a random variable X is defined as

$$M_X(t) = E[e^{tX}]$$

for $t \in \mathbb{R}$ such that $E[e^{tX}] < \infty$.

-> eg: 4.8

For $X \sim \text{Poisson}(\lambda)$ with $\lambda > 0$,

$$M_X(t) = E[e^{tX}] = \sum_{k=0}^{\infty} e^{tk} P(X=k)$$

$$= \sum_{k=0}^{\infty} e^{tk} \frac{\lambda^k}{k!} e^{-\lambda} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} = e^{-\lambda} e^{\lambda e^t}$$

$$= e^{\lambda(e^t - 1)}, \quad t \in \mathbb{R}$$

-> 4.11:

Let X and Y be random variables

(i) If there is an $\varepsilon > 0$ such that $M_X(t)$ and $M_Y(t)$ exist for $t \in (-\varepsilon, \varepsilon)$ and $M_X(t) = M_Y(t)$ for $t \in (-\varepsilon, \varepsilon)$, then X and Y have the same distribution.

(ii) Let $\varepsilon > 0$ be such that $M_X(t)$ exists for all $t \in (-\varepsilon, \varepsilon)$. Then, for all $m \in \mathbb{N}$, $E[|X|^m] < \infty$ and

$$E[X^m] = M_X^{(m)}(0)$$

Stochastics - KVS

→ 4.14 :

Let X and Y be independent random variables. Then,

$$M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$$

for all $t \in \mathbb{R}$, such that $M_X(t)$ & $M_Y(t)$ exists.

→ e.g. 4.15:

Let $X \sim \text{Poisson}(\lambda)$ & $Y \sim \text{Poisson}(\mu)$ with $\lambda, \mu > 0$ be independent. Then,

$$M_{X+Y}(t) = M_X(t) M_Y(t) = e^{\lambda(e^t-1)} e^{\mu(e^t-1)}$$
$$= e^{(\lambda+\mu)(e^t-1)}$$

Stochastics - KVB - Ex 8

①

For $n \in \mathbb{N}$, $p \in (0,1)$ let $X \sim \text{Binom}(n,p)$ and $Y \sim \text{Ber}(p)$ be two independent random variables.

a) Determine $P(X+Y=k)$

$X \sim \text{Binom}(n,p) \Rightarrow P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$

$Y \sim \text{Ber}(p) \Rightarrow Y \sim \text{Binom}(1,p)$

$$P(X+Y=m) = \sum_{k=-\infty}^{\infty} P(X=m-k)\,P(Y=k) \quad \leftarrow k \notin \{0,1\} = 0$$

$$= \sum_{k=0}^{1} P(X=m-k)\,P(Y=k) = P(X=m)P(Y=0) + P(X=m-1)P(Y=1)$$

$$= \binom{n}{m} p^m (1-p)^{n-m} \cdot \binom{1}{0} p^0 (1-p)^{1-0} + \binom{n}{m-1} p^{m-1} (1-p)^{n-(m-1)} \cdot \binom{1}{1} p^1 (1-p)^{1-1}$$

$$= \binom{n}{m} p^m (1-p)^{n-m} \cdot (1-p) + \binom{n}{m-1} p^{m-1} (1-p)^{n-m+1} \cdot p$$

$$= p^m (1-p)^{n-m+1} \left(\underbrace{\binom{n}{m} + \binom{n}{m-1}}_{\binom{n+1}{m}} \right) = p^m (1-p)^{n-m+1} \binom{n+1}{m}$$

$\Rightarrow X+Y \sim \text{Binom}(n+1,p)$

Stochastics - HW6 - 6x8

1

b) Calculate the moment generating function of X & Y

$$M_X(t) = E[e^{tX}] = \sum_{k=-\infty}^{\infty} e^{tk} \binom{n}{k} p^k (1-p)^{n-k}$$

$$= \sum_{k=0}^{n} e^{tk} \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} (e^t \cdot p)^k \cdot (1-p)^{n-k}$$

$$= (e^t p + 1 - p)^n$$

Since $Y \sim \text{Bin}(1, p)$

$$M_Y(t) = e^t p + 1 - p$$

c)

$$M_{X+Y}(t) = M_X(t) \cdot M_Y(t) = (pe^t + (1-p))^{n+1}$$

$$\Rightarrow X + Y \sim \text{Bin}(n+1, p)$$

Stochastics - KVS - Ex 8

(2)

Let $\lambda > 0$ and $X \sim Exp(\lambda)$. Show that:

$$M_X(t) = \frac{\lambda}{\lambda - t}$$

$(g(h(x)))' = g'(h(x)) h'(x)$

Compute $E[X^n]$ & $Var[X]$

$$E[X^n] = M_X^{(n)}(0)$$

$$M_X'(t) = \left(\frac{\lambda}{\lambda - t}\right)' = (\lambda \cdot (\lambda - t)^{-1})' = -\lambda \cdot (-1) \cdot (\lambda - t)^{-2} = \frac{\lambda}{(\lambda - t)^2}$$

$$M_X''(t) = \left(\frac{\lambda}{(\lambda - t)^2}\right)' = (\lambda \cdot (\lambda - t)^{-2})' = (-2) \cdot (-1) \cdot \lambda \cdot (\lambda - t)^{-3} = \frac{2! \lambda}{(\lambda - t)^3}$$

$$M_X^{(k)}(t) = \frac{\lambda \cdot k!}{(\lambda - t)^{k+1}}$$

$$E[X^k] = M_X^{(k)}(0) = \frac{\lambda \cdot k!}{\lambda^{k+1}} = \frac{k!}{\lambda^k}$$

$$Var[X] = E[X^2] - E[X]^2 = \frac{1}{\lambda^2}$$

Stochastik – KW 8 – 6×8

3) Consider the following random experiment: You roll a fair die with six sides and a fair die with six sides where three of its six sides show the number one and the remaining two

a) Determine a suitable probability space (Ω, P)

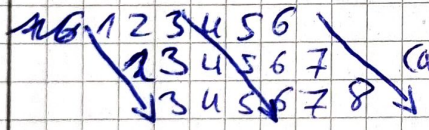
$$\Omega = \{1,..,6\} \times \{1,2\}$$

$$P(A) = |A| \cdot \frac{1}{2} \cdot \frac{1}{6} = \frac{|A|}{12}$$

b) Let A be the event that the die only with 1/2 shows one (1) and B_i the event that the sum of both dies equals i for $i \in \mathbb{N}$. $P(B_i | A)$ for $i \in \mathbb{N}$

$$A = \{(a,1) : a \in \{1,..,6\}\} \Rightarrow |A| = 6$$

$$P(A) = \frac{1}{2}$$



$$P(B_i | A) = \frac{1}{6} \cdot \frac{1}{2}, \text{ for } i \in \{2,..,7\}$$

$$B_i \cap A = \{(a,1) : a \in \{1,..,6\}, a+1 = i\}$$

$1 \geq i > 1$: $|B_i \cap A| = 1 \Rightarrow P(B_i \cap A) = 1/6$

$i = 1$: $|B_1 \cap A| = 0 \Rightarrow P(B_1 \cap A) = 0$

$i > 7$: $|B_i \cap A| = 0 \Rightarrow P(B_i \cap A) = 0$

$$P(B_i | A) = \frac{P(B_i \cap A)}{P(A)} = \begin{cases} \frac{1}{6}, & i \in \{2,..,7\} \\ 0, & \text{else} \end{cases}$$

Stochastik - KVB - 6x8

(3)

c) Let X denote the random variable corresponding to the result of the die with the numbers from one to six and Y the random variable corresponding to the results of the other die. Define X & Y as random variables on your probability space and compute $E[X+Y|A]$

$\omega \in \Omega \quad = (\omega_1, \omega_2) \in \Omega$

$k \in \mathbb{N}$

$X: \omega \in \Omega \to \omega_1$

$Y: \omega \in \Omega \to \omega_2$

$P(X+Y=k|A) = P(B_k|A)$

$$E[X+Y|A] = \sum_{k=1}^{\infty} k \cdot \underbrace{P(X+Y|A)}_{1/6 \text{ for } k=2,\dots,7}$$

$$\equiv \begin{matrix} \\ 0 \end{matrix}$$

$$= \sum_{k=2}^{7} k \cdot \frac{1}{6} = 4,5$$

$$E[\circ|A] = \sum_{k=-\infty}^{\infty} k\, P(\{k\}|A)$$

Stochastik - CVB - Ex 8

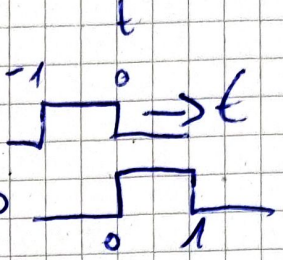
(4)

Let X and Y be two independent random variables which are uniformly distributed on $[0,1]$.

a) Show that the density of $Z = X+Y$:

$$f_Z(x) = \begin{cases} x & , x \in [0,1] \\ 2-x & , x \in (1,2] \\ 0 & , \text{else} \end{cases}$$

$$f_X(x) = \begin{cases} 1 & , x \in [0,1] \\ 0 & , \text{else} \end{cases}$$



$t-s$

$-(s-t)$

$$f_{X+Y}(t) = (f_X * f_Y)(t) = \int_{-\infty}^{\infty} f_X(t-s) f_Y(s)\, ds$$

$$= \int_0^1 1 \cdot f_X(t-s)\, ds$$

$$= \begin{cases} \int_0^t 1 \, ds, & t \in [0,1] \\ \int_{t-1}^1 1 \, ds, & t \in (1,2] \\ 0, & \text{else} \end{cases} = \begin{cases} x, & t \in [0,1] \\ 2-x, & t \in (1,2] \\ 0, & \text{else} \end{cases}$$

$$\int_0^t 1 \, ds = [x]_0^t = t$$

$$\int_t^2 1 \, ds = [x]_t^2 = 2-t$$

Stochastic - KVB - 6x8

(4)

b) Moment generating function Z

$$M_Z(t) = E[e^{tz}] = \int_{-\infty}^{\infty} e^{tz} \cdot f_Z(z)\, dz$$

$$= \int_0^1 e^{tz} \cdot z\, dx + \int_1^2 e^{tz}(2-z)\, dz$$

$$M_X(t) = E[e^{tx}] = \int_{-\infty}^{\infty} e^{tx} f_X(x)\, dx = \int_0^1 e^{tx} dx$$

$$= \begin{cases} \left[\frac{1}{t} e^{tx}\right]_0^1 = \frac{1}{t}(e^t - 1), & t \neq 0 \\ \int_0^1 1\, dx = 1, & t = 0 \end{cases}$$

$$M_{X+Y}(t) = M_X(t) M_Y(t) = M_X(t)^2$$

$$M_Z(t) = \begin{cases} \frac{1}{t^2}(e^t - 1)^2, & t \neq 0 \\ 1, & t = 0 \end{cases}$$

Stochastik - XLIB - Ex 8

(5)

Let T be an $\mathbb{N}_0$-valued random variable such that $E[T] < \infty$ and let $X_1, X_2, \ldots$ be a sq. independent and identically distributed random variables (independent of T, also having finite expectation). Define the random sum

$$Z = \sum_{k=1}^{T} X_k$$

and for $n \in \mathbb{N}_0$ let A_n denotes the event that T equals n Σ $A_n = \{T = n\}$ where you assume that $P(A_n) > 0$ for all $n \in \mathbb{N}_0$

a) Compute $E[Z \mid A_n]$

$$E[Z \mid A_n] = \frac{E[Z \, 1_{A_n}]}{P(A_n)} = \frac{E[1_{A_n} \sum_{k=1}^{T} X_k]}{P(A_n)}$$

$$= \frac{E[1_{\{T=n\}} \sum_{k=1}^{T} X_k]}{P(A_n)} = \frac{E[1_{\{T=n\}} \sum_{k=1}^{n} X_k]}{P(A_n)}$$

$$\overset{\text{indep.}}{=} \frac{E[1_{\{T=n\}}] \sum_{k=1}^{n} E[X_1]}{P(A_n)} = \sum_{k=1}^{n} E[X_1] = n \cdot E[X_1]$$

Stochastik - KVB - 6

(5)

b) Show $E[Z] = E[N]\,E[X]$

$A_n \cap A_m = \emptyset \quad , n \neq m$ ✓

$P(A_n) > 0 \quad , n \in \mathbb{N}_0$ ✓

$\bigcup_{n \in \mathbb{N}_0} A_n = \Omega$ ✓

$$E[X] = \sum_{i \in I} P(B_i)\, E[X \mid B_i]$$

$$E[Z] = \sum_{n=0}^{\infty} P(A_n)\, E[Z \mid A_n]$$

$$= \sum_{n=0}^{\infty} P(A_n) \cdot n \cdot E[X] = E[N] \cdot E[X]$$

($P(A_n) \cdot n$: $E[N]$)

Stochastics - KVB

→ Markov inequality

Let X be a random variable and let $h: [0, \infty) \to [0, \infty)$ be a monotonously non-decreasing function such that $E[h(|X|)] < \infty$. Then, for any $u > 0$ with $h(u) > 0$,

$$P(|X| \geq u) \leq \frac{E[h(|X|)]}{h(u)}$$

→ Chebyshev inequality

For a random variable X such that $E[X^2] < \infty$ and $u > 0$,

$$P(|X - E[X]| \geq u) \leq \frac{Var[X]}{u^2}$$

→ eg: 4.20:

For $N \sim Normal((0,1))$ and $u > 0$, by 4.19

$$P(|N| \geq u) = P(|N - E[N]| \geq u)$$
$$\leq \frac{Var[N]}{u^2} = \frac{1}{u^2}$$

→ Convergence in probability:

A sequence of random variables $(X_n)_{n \in \mathbb{N}}$ converges in probability to a random variable X if

$$\lim_{n \to \infty} P(|X_n - X| \geq \varepsilon) = 0$$

for all $\varepsilon > 0$. We write $X_n \xrightarrow{P} X$ as $n \to \infty$.

An arbitrary collection of random variables is independent if any finite subset of them is independent.

i.i.d = independent and identically distributed

Stochastics - KVB

→ Weak law of large numbers

Let $(X_i)_{i \in \mathbb{N}}$ be a sequence of i.i.d. random variables such that $E[X_1^2] < \infty$ and define $S_n = \sum_{i=1}^{n} X_i$ for $n \in \mathbb{N}$

Then,

$$\frac{S_n}{n} \xrightarrow{P} E[X_1] \quad \text{as } n \to \infty$$

$$\to P\left(\left|\frac{S_n}{n} - E[X_1]\right| \geq \varepsilon\right) \leq \frac{Var[X_1]}{\varepsilon^2 n}$$

→ eg: 4.24:

Let $(X_i)_{i \in \mathbb{N}}$ be i.i.d. $Ber(p)$-distributed random variables with $p \in [0,1]$. Since $E[X_1] = p$ and $E[X_1^2] = p < \infty$ by Theorem 4.22:

$$\frac{1}{n} \sum_{i=1}^{n} X_i \xrightarrow{P} p \quad \text{as } n \to \infty$$

For a random experiment with an event A such that $P(A) = p$ and $X_i = 1_{[A \text{ in } i\text{th repetition}]}$, $i \in \mathbb{N}$

$\frac{1}{n} \sum_{i=1}^{n} X_i$ is the relative frequency of A

Stochastics - KVB

→ Convergence in distribution

A sequence of random variables $(X_n)_{n \in \mathbb{N}}$ converges in distribution to a random variable X if

$$\lim_{n \to \infty} F_{X_n}(t) = F_X(t)$$

for all $t \in \mathbb{R}$ such that F_X is continuous in t.

We write $X_n \xrightarrow{d} X$ as $n \to \infty$.

Φ distribution function of standard normal distribution

→ Central limit theorem:

Let $(X_i)_{i \in \mathbb{N}}$ be a sequence of i.i.d. random variables such that $E[X_1^2] < \infty$ and $Var[X_1] > 0$, $S_n = \sum_{i=1}^{n} X_i$ for $n \in \mathbb{N}$ and $N \sim Normal(0,1)$. Then

$$\lim_{n \to \infty} P\left(\frac{S_n - n E[X_1]}{\sqrt{Var[X_1]}\sqrt{n}} \leq t\right) = \Phi(t)$$

for $t \in \mathbb{R}$ and

$$\frac{S_n - n E[X_1]}{\sqrt{Var[X_1]}\sqrt{n}} \xrightarrow{d} N \quad \text{as } n \to \infty$$

$\to E[S_n] = n E[X_1]$ & $Var[S_n] = n \, Var[X_1]$

$$\to \frac{S_n - n E[X_1]}{\sqrt{Var[X_1]}\sqrt{n}} = \frac{\sqrt{n}}{\sqrt{Var[X_1]}} \left(\frac{S_n}{n} - E[X_1]\right)$$

Stochastics - KVB

→ De Moivre - Laplace Theorem

Let $p \in (0,1)$ and let $Y_n \sim Bi(n,p)$ for $n \in \mathbb{N}$.

Then,

$$\lim_{n \to \infty} P\left(\frac{Y_n - np}{\sqrt{np(1-p)}} \leq \epsilon\right) = \Phi(\epsilon)$$

for $\epsilon \in \mathbb{R}$

$$P(Y_n \leq u) = P\left(\frac{Y_n - np}{\sqrt{np(1-p)}} \leq \frac{u - np}{\sqrt{np(1-p)}}\right) \approx \Phi\left(\frac{u-np}{\sqrt{np(1-p)}}\right)$$

for $u \in \mathbb{R}$

Stochastics – KVB – Ex 9

(1)

For $n \in \mathbb{N}$ let X_n be discretely uniformly distributed on the set $\{1/n, 2/n, \dots, n/n\}$ and let $X \sim [0,1]$

a) Show that the distribution function of X_n is given by

$$P(X_n \leq t) = F_n(t) = \begin{cases} 0 & , \text{for } t < 0 \\ \frac{\lfloor nt \rfloor}{n} & , \text{for } t \in (0,1) \\ 1 & , \text{for } t \geq 1 \end{cases}$$

$$P(X_n = t) = \frac{1}{n} = P(X_n = \frac{j}{n}) = \frac{1}{n} \qquad j \in \{1, \dots, n\}$$

[illegible]

| überalle mit Bedingung !

$$F_n(t) = \sum_{\substack{x \in X(\mathbb{R}) \\ x \leq t}} P(X_n = x) = \sum_{\substack{j=1 \\ \frac{j}{n} \leq t}}^{n} P(X_n = j/n)$$

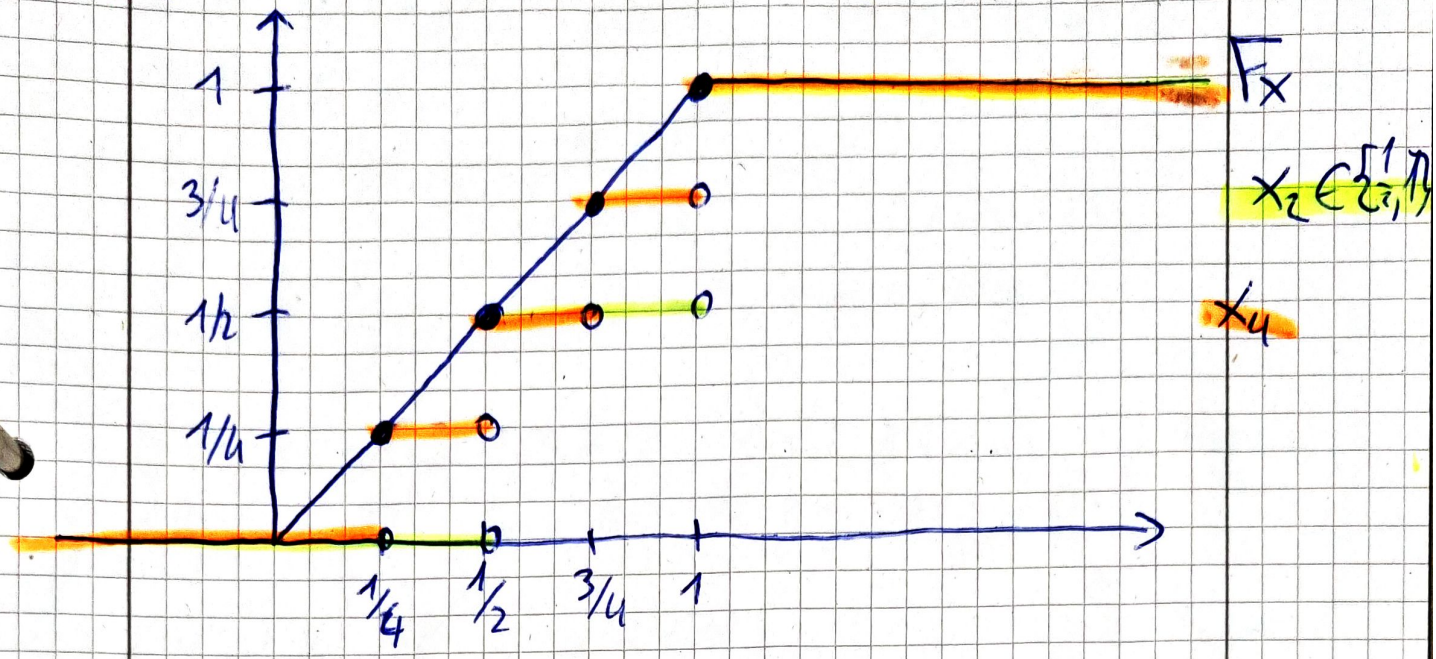
$$= \frac{1}{n} \cdot \left| \{ j \in \{1, \dots, n\} \mid \frac{j}{n} \leq t \} \right|$$

$$= \frac{1}{n} \left| \{ j \in \{1, \dots, n\} \mid j \leq n \cdot t \} \right| = \begin{cases} 0 & , t \leq 0 \\ \frac{\lfloor nt \rfloor}{n} & , t \in (0,1) \\ 1 & , t \geq 1 \end{cases}$$

Stochastic - KVB - 6.9

(1)

b) Sketch the distribution functions of X, X_2 & X_4.



c) Show that $X_n \xrightarrow{d} X$ as $n \to \infty$ where you may use that for all $x \in \mathbb{R}$ it holds that $\frac{[x_n]}{n} \xrightarrow{n\to\infty} X$

$$\lim_{n\to\infty} P(|X_n - X| \geq \varepsilon) = 0$$

$$\lim_{n\to\infty} F_{X_n}(t) \overset{n\to\infty}{=} \begin{cases} 0, & t \leq 0 \\ \lim_{n\to\infty} \frac{[nt]}{n}, & t \in (0,1) \\ 1, & t \geq 1 \end{cases} = \begin{cases} 0, & t \leq 0 \\ t, & t \in (0,1) \\ 1, & t \geq 1 \end{cases}$$

$$\lim_{n\to\infty} \frac{[nt]}{n} \overset{n\to\infty}{=} t$$

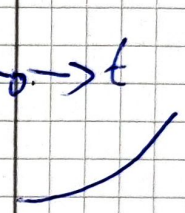
Stochastics - KVB - Ex 9

②

The so-called Erlang distribution with parameters $\lambda > 0$ and $n \in \mathbb{N}$, denoted by $Erl(\lambda, n)$, is given by the density

$$f(x) = \begin{cases} \frac{\lambda^n x^{n-1}}{(n-1)!} e^{-\lambda x} & , x > 0 \\ 0 & , \text{else} \end{cases}$$

(a) For $n \in \mathbb{N}$ and $\lambda > 0$ let $X \sim Erl(\lambda, n)$ and $Y \sim Exp(\lambda)$ be independent. Show that $X + Y \sim Erl(\lambda, n+1)$.



$$f_Y(x) = \begin{cases} \lambda e^{-\lambda x} & , x \geq 0 \\ 0 & , x < 0 \end{cases}$$

$$f_{X+Y}(t) = f_X * f_Y = \int_{-\infty}^{\infty} f_X(t-s) f_Y(s) \, ds$$

$$= \int_0^{\infty} f_X(t-s) \cdot \lambda e^{-\lambda s} \, ds$$

$$= \begin{cases} 0 & , t \leq 0 \\ \int_0^t f_X(t-s) \lambda e^{-\lambda s} \, ds & , t > 0 \end{cases}$$

Stochast - KÜS - Ex 9

②

a) $$\int_0^t f_X(t-s)\,\lambda e^{-\lambda s}\,ds = \int_0^t \frac{\lambda^n (t-s)^{n-1}}{(n-1)!}\,\lambda e^{-\lambda s}\cdot e^{-\lambda(t-s)}\,ds$$

$$= \int_0^t \frac{\lambda^{n+1}}{(n-1)!}\,(t-s)^{n-1}\,e^{-\lambda s-\lambda t+\lambda s}\,ds$$

$$= \frac{\lambda^{n+1}}{(n-1)!}\cdot e^{-\lambda t}\int_0^t (t-s)^{n-1}\,ds \qquad u = t-s$$

$$du = -1\,ds$$

$$= \frac{-\lambda^{n+1}}{(n-1)!}\,e^{-\lambda t}\int_t^0 u^{n-1}\,du = \frac{\lambda^{n+1}\,t^n}{n!}\,e^{-\lambda t}$$

$\rightsquigarrow X+Y \sim Erl(\lambda, n+1)$

Stochastik – KVB – 6x9

②

b) Show for $\lambda > 0$, $n \in \mathbb{N}$ and i.i.d random variables $X_1, \dots, X_n$ with $X_1 \sim Exp(\lambda)$ that

$$\sum_{k=1}^{n} X_k \sim Erl(\lambda, n)$$

IA:

$$f_{X_1+X_2} = \int_{-\infty}^{\infty} f_{X_1}(t-s) f_{X_2}(s)\, ds = \int_0^{\infty} f_{X_1}(t-s)\, \lambda e^{-\lambda s}\, ds$$

$$= \begin{cases} 0, & t \leq 0 \\ \int_0^t f_{X_1}(t-s) \lambda e^{-\lambda s}\, ds \end{cases}$$

$$\int_0^t f_{X_1}(t-s) \lambda e^{-\lambda s}\, ds = \int_0^t \lambda^2 e^{-\lambda s - \lambda t + \lambda s}\, dt$$

$$= \lambda^2 e^{-\lambda t} \int_0^t 1\, dt$$

$$= \lambda^2 e^{-\lambda t} \cdot t$$

$$\rightsquigarrow \sum_{k=1}^{2} X_k \sim Erl(\lambda, 2)$$

IS:

$n-1 \to n$

siehe a)

$$\Rightarrow \sum_{k=1}^{n} X_k \sim Erl(\lambda, n)$$

Stochastics - KVB - 6.x.9

(2)

a) Let $Y_n \sim Erl(\lambda, n)$ for $\lambda > 0$ and $n \in \mathbb{N}_0$.
Show for $N \sim Normal(0,1)$ that

$$\frac{Y_n - n/\lambda}{\sqrt{n}/\lambda} \xrightarrow{d} N \qquad \text{as } n \to \infty$$

Normal:

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad x \in \mathbb{R}$$

$$\Phi(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{y} e^{-x^2/2}\, dx$$

Y_n has the same distribution as $\sum_{k=1}^{n} X_k$ for $X_1, X_2, \ldots \overset{i.i.d}{\sim} Exp(\lambda)$

$$E[X_k] = \frac{1}{\lambda} \; ; \; Var[X_k] = \frac{1}{\lambda^2}$$

$$\frac{\sum_{k=1}^{n} X_k - \frac{n}{\lambda}}{\sqrt{n}/\lambda} \xrightarrow{d} N$$

In die Form bringen

$$\hookrightarrow = \frac{\sum_{k=1}^{n} X_k - n \cdot E[X_1]}{\sqrt{n} \cdot \sqrt{Var[X_1]}} \xrightarrow{d} N$$

Stochastic - R.V.B - Exp

(3)

Suppose that you roll a pair die 900 times and let X denote the random variable which counts how often the die shows a six

a) Use the Markov inequality with $h(x)=x$ to bound $P(X \geq 200)$

$$P(X=k) = \binom{900}{k}\left(\frac{1}{6}\right)^k\left(\frac{5}{6}\right)^{900-k}$$

$$P(|X| \geq 200) \leq \frac{E[|X|]}{200}$$

$$E[h(x)] = E[X] = \sum_{k=0}^{900} k \cdot P(X=k)$$

$$= \sum_{k=0}^{900} k \cdot \frac{900!}{k!(900-k)!}\left(\frac{1}{6}\right)^k\left(\frac{5}{6}\right)^{900-k} = \sum_{k=1}^{900} \frac{900!}{(k-1)!(900-k)!}\left(\frac{1}{6}\right)^k\left(\frac{5}{6}\right)^{900-k}$$

$$\ell = k-1 \Rightarrow k = \ell+1$$

$$= \sum_{\ell=0}^{899} \frac{900 \cdot 899!}{\ell!(899-\ell)!} \cdot \frac{1}{6}\cdot\left(\frac{1}{6}\right)^{\ell}\left(\frac{5}{6}\right)^{899-\ell}$$

$$= \frac{900}{6}\left(\frac{1}{6}+\frac{5}{6}\right)^{899} = \underline{150}$$

$$P(|X| \geq 200) \leq \frac{150}{200} \qquad P(|X| \geq 160) \leq \frac{150}{160}$$

Stochastyka - KRB - 6.05

③

b) Chebyshev inequality

$$P(|X - E[X]| \geq u) \leq \frac{Var[X]}{u^2}$$

$$Var[X] = 900 \cdot \frac{1}{6} \cdot \frac{5}{6} = 125$$

$P(X \geq 200) = P(|X| \geq 200)$

$P(X \geq 160) \leq P(|X| \geq 160)$

$$P(|X - 150| \geq 200 - E[X]) \leq \frac{Var(X)}{50^2} = 0{,}05$$

$$P(|X - 150| \geq 10) \leq \frac{Var(X)}{10^2} = 1{,}25$$

$P(X \geq 200) = P(X - E[X] \geq 200 - E[X])$)

(3)

c) Use normal approximation.

$$P(X \geq 200) = 1 - P(X \leq 199) = 1 - P\left(\frac{X-150}{\sqrt{125}} \leq \frac{199-150}{\sqrt{125}}\right) = 1 - P\left(\frac{X-150}{\sqrt{125}} \leq \frac{49}{\sqrt{125}}\right)$$

$$\approx 1 - \phi(4{,}38) \approx 6 \cdot 10^{-6}$$

$$P(X \geq 160) = 1 - P(X \leq 159) \approx 1 - \phi(0{,}80) \approx 0{,}212$$

$$P(Y_n \leq h) = P\left(\frac{Y_n - np}{\sqrt{np(1-p)}} \leq \frac{u-np}{\sqrt{np(1-p)}}\right) \approx \phi\left(\frac{u-np}{\sqrt{np(1-p)}}\right) \Big|_0$$

Stochastik IC VB - 6x9

(4)

a) For $n \in \mathbb{N}$ let $X_n \sim Exp(n)$. $\to (0, \infty)$

Show $\forall \varepsilon > 0 : P(|X_n - 0| > \varepsilon) \to 0$

Let $\varepsilon > 0$, it holds

$$P(|X_n - 0| > \varepsilon) \overset{\text{sea}}{=} P(|X_n| > \varepsilon) = P(X_n > \varepsilon)$$

$$= 1 - P(X_n \leq \varepsilon)$$

$$= 1 - (1 - e^{-n\varepsilon}) = e^{-n\varepsilon} \overset{n \to \infty}{\to} 0$$ ▨

b) For $p \in (0,1)$ and a sequence $(Y_n)_{n \in \mathbb{N}}$ of i.i.d. random variables with $Y_1 \sim Ber(p)$ we define $X_n = \frac{1}{n} \sum_{i=1}^{n} Y_i$ for $n \in \mathbb{N}$. Show that $X_n \xrightarrow{P} 0$ as $n \to \infty$

$$P(|X_n - 0| > \varepsilon) = P(X_n > \varepsilon) = P\left(\frac{1}{n} \sum_{i=1}^{n} Y_i > \varepsilon\right) = \begin{cases} 0, & \varepsilon \geq 1 \\ ?, & \varepsilon < 1 \end{cases}$$

Let now $0 < \varepsilon < 1$:

$$P\left(\frac{1}{n} \sum_{i=1}^{n} Y_i > \varepsilon\right) = P\left(\prod_{i=1}^{n} Y_i = 1\right) = P(Y_1 = 1, Y_2 = 1, \dots, Y_n = 1)$$

$$= P(Y_1 = 1) P(Y_2 = 1) \cdot \dots \cdot P(Y_n = 1) = \underbrace{p^n}_{0 < p < 1} \overset{n \to \infty}{\to} 0$$